

REPOSITORY HARDENING REPORT

512 / CVS Evidence-Sidecar

Pre-Seal Hardening Pass — Execution Boundary Integrity

Date	02 April 2026
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Pass type	Execution boundary hardening — commit path ownership, non-conformant pattern elimination, primitive boundary declaration, pre-seal status marking
Repos affected	2 — 512 (JonathanMastersWatson/512), CVS Evidence-Sidecar (Evidence-Sidecar/CVS)
Total files changed	17 files across two repositories (9 new, 8 modified)
Canon docs updated	512_ARCHITECTURE v2.9 → v3.0 512_IMPLEMENTATION v1.10 → v2.0
Seal status	Pre-seal hardening phase. Archive hashes to be computed after next full pass.

1. Executive Summary

This report documents the hardening pass conducted on 02 April 2026 across two repositories: 512 (JonathanMastersWatson/512) and CVS Evidence-Sidecar (Evidence-Sidecar/CVS). This pass was driven by a pre-hardening note identifying a consistent implementation failure pattern: systems introducing a structural gap between decision and execution, treating 512 as a pre-check rather than the owner of the only commit path.

The core constraint hardened throughout this pass:

There exists exactly one path to irreversible state change, and that path is owned and resolved by the gate. If any execution path exists outside of gate control, the system is non-conformant.

17 files were changed across both repositories (9 new, 8 modified). Both repositories are now marked as pre-seal hardening phase. Canon documents 512_ARCHITECTURE and 512_IMPLEMENTATION were updated to v3.0 and v2.0 respectively and delivered as MD and PDF. CVS canon documents are deferred to the next pass.

2. What Changed — by Repository

2.1 512 Repository — 8 files (3 new, 5 modified)

File	Action	What it addresses
PRE_HARDENING_NOTICE.md	New	Pre-seal status marker. Communicates to implementers: architectural constraints directionally stable; language still being tightened; constraints clarify but never relax.
LIVING_DOCUMENTS.md	New	Identifies dynamic vs frozen documents. Distinguishes living docs (architecture, ops references) from frozen artifacts (kernel, invariants, genesis, proofs).
PRIMITIVE_BOUNDARY.md	New	Authoritative one-page boundary declaration: 512 defines constraint behaviour at the execution boundary; does not define architecture, orchestration, deployment, or implementation patterns; derivative responsibility rests with implementing party.
README.md	Modified	Pre-hardening callout added. PRIMITIVE_BOUNDARY.md and LIVING_DOCUMENTS.md added to Reference section and Status section.
ANTI_DRIFT.md	Modified	Pattern 7 (Bypass Accumulation) strengthened: disqualifying condition is existence of bypass path, not frequency of use. Procedural controls explicitly insufficient. Pattern 8 (Pre-Check Positioning) added: names the architectural drift pattern where evaluation runs upstream of an independently operable execution surface.
FAILURE_MODES.md	Modified	Two language conflicts fixed. §1: 'only witnesses that an execution occurred' (old observation framing) replaced with accurate statement of evaluation scope. §3: 'does not block or permit actions' (contradicts allow/deny model) replaced with accurate scope — 512 produces allow/deny/gap at boundary; post-hoc consequences are out of scope.

512-ops/COMMIT_BOUNDARY_REFERENCE.md	Modified	§8 Commit Path Ownership added: single-path requirement defined structurally; procedural controls explicitly insufficient; existence of bypass path is the disqualifying condition. §9 Non-Conformant Patterns added: five patterns explicitly labelled and diagrammed.
512-ops/COMPLIANCE_CHECKLIST.md	Modified	Commit Path Properties section added: four new checklist items covering single-path requirement, structural prerequisite requirement, procedural controls insufficiency, and pre-check architecture non-conformance.

2.2 CVS Repository — 7 files (3 new, 4 modified)

File	Action	What it addresses
PRE_HARDENING_NOTICE.md	New	Pre-seal status marker scoped to CVS evidence formation model. Same structure as 512 equivalent.
LIVING_DOCUMENTS.md	New	CVS equivalent dynamic/frozen document registry. Living: CVS_ARCHITECTURE, CVS_IMPLEMENTATION, ANTI_DRIFT.md, CONFORMANCE.md. Frozen: CANON_HASHES.md, LEGAL_NOTICE.md, LICENSE, NOTICE, UPSTREAM.md, PRIMITIVE_BOUNDARY.md.
PRIMITIVE_BOUNDARY.md	New	CVS-scoped primitive boundary declaration: CVS defines evidence formation at the execution boundary; does not define constraint behaviour, system architecture, orchestration, deployment models, or implementation patterns.
README.md	Modified	Pre-hardening callout added near top. Reference section added at end: PRIMITIVE_BOUNDARY.md, LIVING_DOCUMENTS.md, PRE_HARDENING_NOTICE.md. Status section updated.
ANTI_DRIFT.md	Modified	Pre-hardening notice callout added. No content changed — all 15 sections, language reference table, and anti-drift principle unchanged.
CONFORMANCE.md	Modified	Pre-hardening notice added. Primitive Boundary section added (derivative responsibility). Language fixes per ANTI_DRIFT.md §13: 'compliance system' → 'conformance system'; 'regulatory compliance' → 'regulatory conformance'; 'logged explicitly' → 'recorded explicitly'. Checklist row 8.1 added: primitive boundary respected. PRIMITIVE_BOUNDARY.md added to Relationship to Other Documents.
00_INTENT/NON_GOALS.md	Modified	Language fix per ANTI_DRIFT.md §13: 'non-compliant with this architecture' → 'non-conformant with this architecture'; 'blocking compliance gates' → 'blocking conformance gates'. Pre-hardening notice added. Truncated final sentence restored.

2.3 Canon Documents Updated

Document	Version	Key changes
512_ARCHITECTURE	v2.9 → v3.0 (major)	New §2.5: Commit Gate Owns the Only Commit Path — defines commit path exclusivity as structural property. Step 4 renamed from Execution Handoff to Commit Authorisation Signal. Commit path exclusivity added to §8.1 Observable Properties and §8.2 Disqualifying Properties. Verification row added to §8.4.
512_IMPLEMENTATION	v1.10 → v2.0 (major)	New §1.5: Commit Path Ownership Is Non-Bypassable — structural requirement, conformant model diagram, MUST/MUST NOT language. New §1.6: Non-Conformant Execution Patterns — five patterns labelled and diagrammed (API handoff, queue handoff, broker handoff, pre-check positioning, parallel/fallback path). Step 4 diagram and text renamed. Two new prohibited behaviours in §11.2. Two new verification tests in §11.4.

3. Drift Vectors Closed

These are the specific misinterpretation surfaces identified and hardened. Each represents a place where an implementer, vendor, or enterprise could have built a non-conformant system while claiming conformance.

Drift vector	How it manifests	Where hardened
Pre-check positioning	Gate positioned upstream of independently operable execution surface; evaluation result is advisory not structural	512/ANTI_DRIFT.md Pattern 8; 512_ARCHITECTURE §2.5; 512_IMPLEMENTATION §1.5, §1.6; COMMIT_BOUNDARY_REFERENCE.md §8, §9
Bypass accumulation	Correctly positioned gate hollowed out by exception paths accumulating over time	512/ANTI_DRIFT.md Pattern 7 (strengthened); disqualifying condition now stated as existence of path, not frequency of use
Procedural bypass controls	Access policies and documented prohibitions substituted for structural elimination of bypass paths	COMMIT_BOUNDARY_REFERENCE.md §8; COMPLIANCE_CHECKLIST.md Commit Path Properties; 512_IMPLEMENTATION §1.5
Decision-execution separation	Gate evaluates and signals; downstream component applies signal to execution surface independently	512_ARCHITECTURE §2.5, §3.3 Step 4; 512_IMPLEMENTATION §1.5, §1.6 Pattern A/B/C
512 as observation layer	'Only witnesses execution' framing implying 512 is passive observer	FAILURE_MODES.md §1 language corrected
512 blocking/permitting framing contradiction	Stating 512 'does not block or permit actions' while also defining allow/deny output	FAILURE_MODES.md §3 language corrected
CVS language drift	'Non-compliant', 'compliance system', 'compliance gates' — prohibited terms per ANTI_DRIFT.md §13	CVS/NON_GOALS.md, CVS/CONFORMANCE.md — all instances corrected to non-conformant/conformance
Primitive scope ambiguity	Implementing party treating CVS or 512 primitives as defining architecture, orchestration, or deployment	PRIMITIVE_BOUNDARY.md (both repos); CONFORMANCE.md Primitive Boundary section

4. Core Constraint Established This Pass

4.1 Commit Path Ownership

The central architectural constraint hardened in this pass — stated for the first time in explicit normative language across both repos:

A Commit Gate is defined by commit path exclusivity: there exists exactly one path to irreversible state change, and that path does not open without the gate's authorisation signal.

This property separates enforcement from observation. A mechanism that evaluates proposals correctly but permits the commit path to operate independently of its signal is a monitoring system — not a Commit Gate.

4.2 Non-Conformant Patterns Explicitly Named

Five implementation patterns are now explicitly labelled NON-CONFORMANT in both the 512 repo (COMMIT_BOUNDARY_REFERENCE.md §9) and 512_IMPLEMENTATION (§1.6):

- Pattern A — Evaluation result handed off to an API layer before DB write
- Pattern B — Evaluation result handed off to a message queue
- Pattern C — Evaluation result handed off to a broker or runtime
- Pattern D — Pre-check positioning (evaluation upstream of independently operable execution surface)
- Pattern E — Parallel or fallback execution path

Each pattern is diagrammed in ASCII and annotated. A developer reading either document cannot implement these patterns and claim conformance.

4.3 Primitive Boundary Declared

Both repos now carry PRIMITIVE_BOUNDARY.md — a short, authoritative, frozen document stating what each primitive defines and what derivative responsibility means:

- 512: defines constraint behaviour at the execution boundary only
- CVS: defines evidence formation at the execution boundary only
- Neither defines architecture, orchestration, deployment models, or implementation patterns
- Any system built using either primitive is a derivative implementation
- Responsibility for design, deployment, and operation remains entirely with the implementing party

5. Language Hardening

Several language violations were identified and corrected across both repos. All corrections follow the substitution table in CVS/ANTI_DRIFT.md §13.

File	Old language	Corrected language
FAILURE_MODES.md §1	512 only witnesses that an execution occurred	512 evaluates whether a proposed action satisfies declared constraints — does not determine whether those constraints are correct or whether declared intent is truthful
FAILURE_MODES.md §3	512 does not block actions / permit actions	512 produces allow/deny/gap at boundary; consequences beyond the boundary decision are out of scope
CVS/NON_GOALS.md	non-compliant with this architecture	non-conformant with this architecture
CVS/NON_GOALS.md	blocking compliance gates	blocking conformance gates
CVS/CONFORMANCE.md	a compliance system	a conformance system
CVS/CONFORMANCE.md	regulatory compliance	regulatory conformance
CVS/CONFORMANCE.md	logged explicitly (State 2)	recorded explicitly

6. Seal Status

Both repositories are in pre-seal hardening phase as of this pass. PRE_HARDENING_NOTICE.md has been added to both repo roots and referenced from README.md.

Pre-seal status means:

- Architectural constraints are directionally stable
- Language is still being tightened
- Constraints may be clarified but will not be relaxed
- Non-conformant patterns identified are final — they will remain non-conformant in sealed versions
- Archive hashes will be computed after the next full pass

Archive hashes from the previous pass (24 March 2026) remain valid for the repos as they existed at that date. The current pass has not been sealed. Sealing will occur after CVS canon documents are updated and the full cross-document audit is complete.

Repo	Status
512	Pre-seal hardening — PRE_HARDENING_NOTICE.md committed
CVS Evidence-Sidecar	Pre-seal hardening — PRE_HARDENING_NOTICE.md committed
Constraint-Architecture	Not touched this pass — seal status unchanged from March 2026

7. What Remains

The following items were identified during this pass and are deferred to the next hardening pass:

Item	Notes
CVS_ARCHITECTURE and CVS_IMPLEMENTATION hardening pass	Canon documents not yet updated this pass — architecture and implementation hardening deferred to next pass
CANON_HASHES.md resealing	Deferred until after next repo hardening pass — both repos still in pre-seal phase
STYLE_GUIDE updates	Pending — no changes made this pass
Constraint-Architecture repo pass	Not touched this pass — pending
Legal review package construction	Four placeholder files for counsel briefing — pending
REFERENCE_FLOW.md completion	Still placeholder — low priority
Cross-document version reference audit	512_IMPLEMENTATION references to architecture version updated; full audit deferred
ANTI_DRIFT.md §15 enforcement language fix (CVS)	Minor fix noted — deferred

8. For Builders — What This Means

If you are building against 512 or CVS, the following files are your primary references after this hardening pass:

512 — Start Here

- 512-core/KERNEL/512-kernel.txt.txt — the 7 invariant statements (the authority)
- 512-core/KERNEL/INVARIANTS.md — each invariant defined precisely
- 512-ops/COMMIT_BOUNDARY_REFERENCE.md — where is the boundary, what crosses it, what comes out, §8-9 for conformant and non-conformant patterns
- ANTI_DRIFT.md — what 512 is not, how implementations drift, including the new Pattern 8
- PRIMITIVE_BOUNDARY.md — what 512 defines and what it does not

CVS — Start Here

- CONFORMANCE.md — what CVS-Conforming means, binary test checklist, primitive boundary
- ANTI_DRIFT.md — layer separation, fixed observation model, prohibited patterns
- PRIMITIVE_BOUNDARY.md — what CVS defines and what it does not
- VERIFICATION_PROTOCOL.md — how to verify evidence as a third party
- 02_EVIDENCE_MODEL/PROOF_OBJECT_INTEGRITY.md — what a Proof Object must contain

Critical Principle

If evaluation can occur without controlling the commit path, enforcement does not exist. The system has a monitoring layer attached to an uncontrolled execution surface. That is not a Commit Gate.

9. Closing Statement

This pass addresses the most common real-world failure mode identified in builder feedback: systems that evaluate constraints correctly but position that evaluation upstream of an independently operable execution surface. Every place where this gap could be introduced — or defended by a vendor — has been closed.

The pre-check architecture pattern, the parallel path pattern, the bypass accumulation pattern, and the procedural-controls-as-substitute pattern are now explicitly named, diagrammed, and labelled NON-CONFORMANT in both canonical repositories. The commit path ownership property is now a named, verifiable, testable architectural property in both the architecture and implementation documents.

Both repositories are in pre-seal hardening state. The next pass will address CVS canon documents and complete the cross-document audit.

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